Shivam Pandey

+91-9136253831 | shivamrajeshpandey9136@gmail.com | LinkedIn | GitHub | Portfolio

Final-year Computer Science and Engineering (AI/ML) student who builds, evaluates and deploys machine learning and LLM systems end to end. Built a RAG app with hybrid retrieval and reranking (90.0% strict Recall@5) and a leakage-audited fraud-detection ensemble with TreeSHAP explanations (F1 0.89 on a hold-out split). Works across Python, XGBoost, FastAPI, PostgreSQL/pgvector and LLM tooling.

Education

Viva Institute of Technology, University of Mumbai, Mumbai *2027*
B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) | CGPI: 8.82/10

Technical Skills

- **ML & Data:** XGBoost, scikit-learn, Isolation Forest, TreeSHAP, SMOTE, NetworkX, Pandas, NumPy
- **GenAI & LLMs:** RAG, hybrid search (BM25 + vector), reranking, Sentence Transformers, Groq, Ollama, OCR
- **Languages:** Python, SQL, JavaScript
- **Backend:** FastAPI, Flask, REST APIs, Gunicorn
- **Frontend:** HTML, CSS, Tailwind CSS, Jinja2
- **Databases:** PostgreSQL, pgvector, MongoDB
- **Tools & Platforms:** Git, GitHub, Docker, Render, Jupyter

Projects

Fund Flow Fraud Detection | Python, XGBoost, Isolation Forest, TreeSHAP, NetworkX GitHub | Live Demo

- Trained an AUC-weighted ensemble (Isolation Forest + 2 XGBoost, class weighting and SMOTE on one model) on PaySim (6.36M transactions, 0.13% fraud); hold-out F1 0.89 (95% CI 0.87–0.90), PR-AUC 0.95.
- Ran a TreeSHAP leakage audit that exposed a simulator artifact and velocity features counting the current transaction; removed both (PR-AUC 0.998 → 0.97), then re-tuned weights and thresholds on a tune split.
- Added top-5 TreeSHAP reasons and NetworkX graph patterns (round-trip, structuring, fan-in/out, 24-hour windows) that can only raise the ML score; deployed on Render with CSV batch jobs and PDF reports.
- Built a CSV pipeline with automatic column mapping and a pause-and-confirm step for missing fields; shrank a 70 MB account lookup to 4 MB to fit a 512 MB free instance.

DocQuery | Python, pgvector, Sentence Transformers, BM25, Ollama/Groq, FastAPI GitHub

- Built a RAG system over uploaded PDFs with streamed, page-cited answers; FastAPI, PostgreSQL/pgvector, Ollama/Groq LLMs, Docker, per-chat isolation.
- Designed hybrid retrieval (vector + BM25, RRF, cross-encoder reranker): Recall@5 76.7% → 90.0%, Recall@10 86.7% → 96.7% vs dense-only (30 questions, strict source+page).
- Engineered the ingestion pipeline (OCR fallback, table extraction, vision-LLM figure descriptions) plus query routing, follow-up rewriting and a reranker-score threshold that returns "not found" instead of guessing.
- Used a per-chat B-tree index instead of HNSW, since an approximate index filtered by chat can return fewer than k results; keyed rank fusion on a full-text hash so chunks sharing a prefix are not merged.

SkillBridge: Career & Learning Platform | Python, Groq LLM, Flask, MongoDB GitHub | Live Demo

- Built and deployed a platform with Groq-generated staged learning roadmaps, an AI chatbot and mock interviews whose difficulty adapts to the user's scores.
- Bounded LLM cost and data exposure: per-account daily limits (50 chats / 10 roadmaps / 5 interviews), API key failover, and only 10 allow-listed profile fields (links and emails stripped) sent to the model.
- Shipped resume (19 templates) and portfolio builders, communities, and admin-moderated uploads feeding an atomic XP/leaderboard system; 34 Python tests + 33 browser checks in GitHub Actions.

Certifications

- **Python for Data Science**, NPTEL (IIT Madras), Elite *2025*
  Learned: NumPy and Pandas, EDA with matplotlib and seaborn, regression and classification case studies.
- **AI/ML Course**, Apna College *2026*
  Learned: supervised and unsupervised learning, precision/recall/F1, neural networks, NLP, RAG, LLM agents.